$\mathrm{Tm}\,(A \Rightarrow B) \cong \mathrm{Tm}\,A \to \mathrm{Tm}\,B$

String $\to \mathrm{Tm}\,B$

$\mathrm{Tm}\,(A+B) \neq \mathrm{Tm}\,A + \mathrm{Tm}\,B$ : f

g

← ok

nem ok →

f (L, a) := inl a

f (L2 b) := inr b

$\mathrm{Ind}\,(\lambda X.\ T + A \times X)$

$A = \underbrace{T + \ldots + T}_{128\times}$

| típus | meta |
|---|---|
| Nat | $\mathbb{N}$ |
| Bool | $\mathbb{Z}$ |
| ⇒ | → |
| inl | $\iota_1$ |
| inr | $\iota_2$ |

$g = \mathrm{case}\ \{A\}\{B\}\{?\}\ (?)\ (!)$

$\mathrm{case}\ \{A\ B\ (C : T_\gamma)\} \to (\mathrm{Tm}\,A \to \mathrm{Tm}\,C) \to (\mathrm{Tm}\,B \to \mathrm{Tm}\,C) \to \mathrm{Tm}\,(A+B) \to \mathrm{Tm}\,C$

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Koinduktív — David Turner — Elementary strong functional prog.

- szerver, op. rendszer, interaktív
- kommunikációs típus = protokoll – ortézisek
- gépek, Turing-gép, valós számok
- folyam = végtelen lista

dualis

Ezüst: destr ↓ generátor $A \to G\,A$ produktív

Arany: konstr ↓ iterátor $\mathrm{Ind}\,F \to A$ terminál

---

Stream : $T_\gamma \to T_\gamma$

head : $\mathrm{Tm}\,(\mathrm{Stream}\,A) \to \mathrm{Tm}\,A$

tail : $\mathrm{Tm}\,(\mathrm{Stream}\,A) \to \mathrm{Tm}\,(\mathrm{Stream}\,A)$

gen $(\mathrm{Tm}\,C \to \mathrm{Tm}\,A) \to (\mathrm{Tm}\,C \to \mathrm{Tm}\,C) \to \mathrm{Tm}\,C \to \mathrm{Tm}\,(\mathrm{Stream}\,A)$

C = mag típusa

$\mathrm{Stream}\beta_1$ : head (gen h t c) = h c

$\mathrm{Stream}\beta_2$ : tail (gen h t c) = gen h t (t c)

xs : $\mathrm{Tm}\,(\mathrm{Stream}\,A)$

head xs : $\mathrm{Tm}\,A$

head (tail xs) : $\mathrm{Tm}\,A$

head (tail (tail xs)) : $\mathrm{Tm}\,1$

ns = 0, 1, 2, 3, ——

ns = gen (λi.i) suc zero

head ns = head (gen (λi.i) suc zero) = (λi.i) zero = zero

head (tail ns) =

head (tail (gen (λi.i) suc zero)) $=_{\beta_2}$

head (gen (λi.i) suc (suc zero)) $=_{\beta_1}$

(λi.i) (suc zero) = suc zero

$F : T_\gamma \to T_\gamma$

$\mathrm{Ind}\,F : T_\gamma$ — $\mathrm{Coind}\,F : T_\gamma$

$\mathrm{Tm}\,(\mathrm{Ind}_F) \xrightarrow{\cong} \mathrm{Tm}\,(P\ \mathrm{Ind}_F)$ — $\mathrm{Tm}\,(\mathrm{Coind}_F) \cong \mathrm{Tm}\,(F\ \mathrm{Coind}_F)$

$\mathrm{Ind}\,F = \mathrm{Coind}\,F$

leglassabb fixpont — legnagyobb

boring = gen (λi.i) (λi.i) zero = (0, 0, 0, 0, 0, —

map : (f : Tm A → Tm B) → Tm (Stream A) → Tm (Stream B)

map f ns = gen (f ∘ head) (tail) as

(λ as. f (head as)) : Tm (Stream A) → Tm B ; : Tm (Stream A) → Tm (Stream A)

map suc (0, 1, 2, 3, — ) = (1, 2, 3, 4, — )

map suc (1, 2, 5, 0, 2, 2, — = (2, 3, 6, 1, 3, 3, —

---

filter : (Tm A → Tm Bool) → Tm (List A) → Tm (List A)

Stream — Stream

---

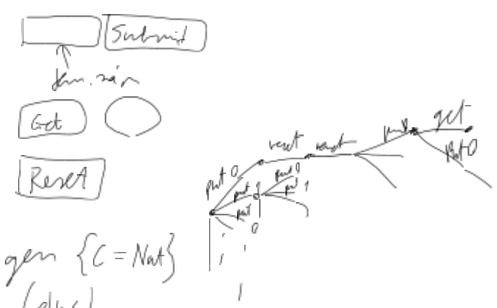
Mach : $T_\gamma$

put : Tm Mach → Tm Nat → Tm Mach

get : Tm Mach → Tm Nat

press : Tm Mach → Tm Mach

gen : (Tm C → Tm Nat → Tm C) → (Tm C → Tm Nat) → (Tm C → Tm C) → Tm C → Tm Mach







sum = gen {C = Nat} (plus) (λi.i) (λ_.zero) zero

maxs = gen {C = Nat} max (λi.i) (λ_.zero) zero

| FX | $\mathrm{Ind}_F$ | $\mathrm{Coind}_F$ | def |
|---|---|---|---|
| T + X | Nat | Conat | Conat → T + Conat |
| T + A × X | $\mathrm{List}_A$ | $\mathrm{Glist}_A$ | |
| A × X | inver. on A × L → L | $\mathrm{Stream}_A$ | |
| X | inver. on L → L | def L → L ≅ T | |
| T + X × X | kin. fa | def L → T + L × L | |

0 := gen {T} (inl) tt

∞ := —

Conat → Nat

$\mathrm{Glist}_A \to T + A \times \mathrm{Glist}_A$

Coind : (F : $T_\gamma \to T_\gamma$) → {Pos F} → $T_\gamma$

def $\mathrm{Tm}\,(\mathrm{Coind}_F) \to \mathrm{Tm}\,(F\ \mathrm{Coind}_F)$

gen : $(\mathrm{Tm}\,C \to \mathrm{Tm}\,(FC)) \to \mathrm{Tm}\,C \to \mathrm{Tm}\,(\mathrm{Coind}_F)$

$\mathrm{Coind}\beta$ : def (gen f c) = $\mathrm{map}_F$ (gen f) (f c)

: Tm (Coind F) ; : Tm (F (Coind F))

$\mathrm{Stream}_A = \mathrm{Coind}\ (\lambda X.\ A \times X)$

def = head + tail

$\mathrm{Tm}\,(\mathrm{Stream}_A) \to \mathrm{Tm}\,(A \times \mathrm{Stream}_A)$

$\mathrm{map}_F$ : (A → B) → F A → F B

$\mathrm{map}_F$ f c : Tm (F C)

(C → Coind) → F C → F Coind

vmi = gen (λ_.zero) (λi.i) tt

fib = 0, 1, 1, 2, 3, 5, 8, 13, —

= gen {C = Nat × Nat} fst (λw. (snd w, fst w + snd w)) (0, 1)